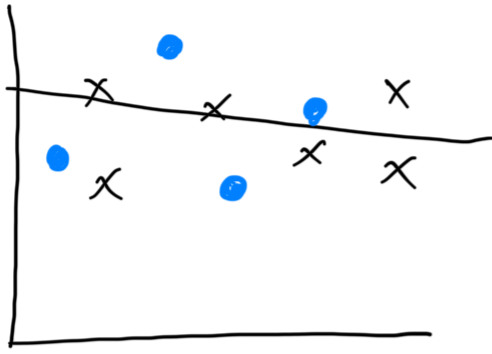
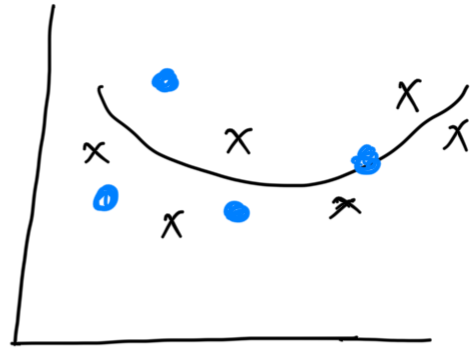


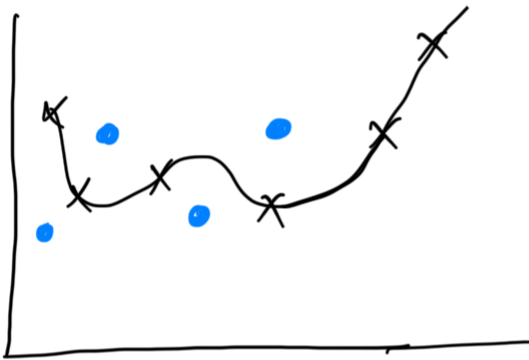
Bias Variance Trade-off.



m_1



m_2



m_3

x - training set

• - test set

Bias - It refers to the gap between predicted values & actual values.

In case of high bias, predictions are likely to be skewed in a certain direction away from actual values.

Variance - It describes how scattered the predicted values are

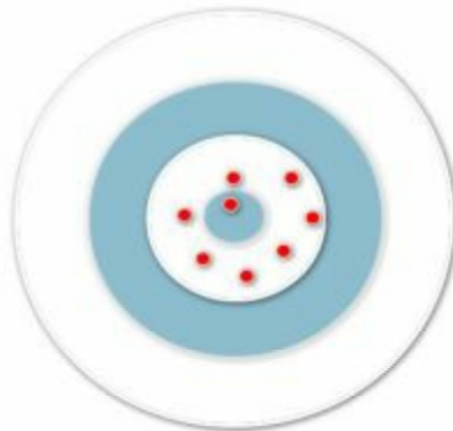
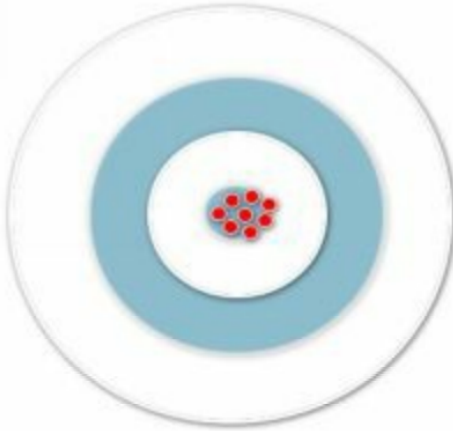
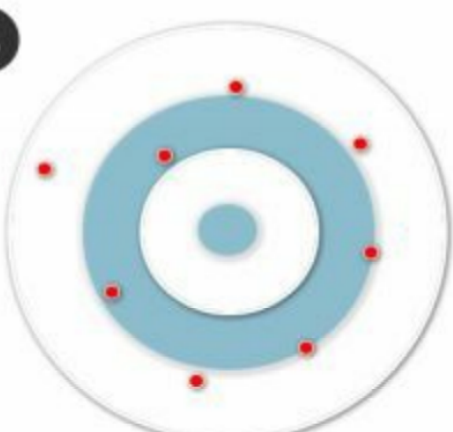
1**Low Variance****2****High Variance****Low
Bias****3****High
Bias****4**

Figure 2: Shooting targets used to represent bias and variance

Bull's eye, perfectly predicts the correct value of model.

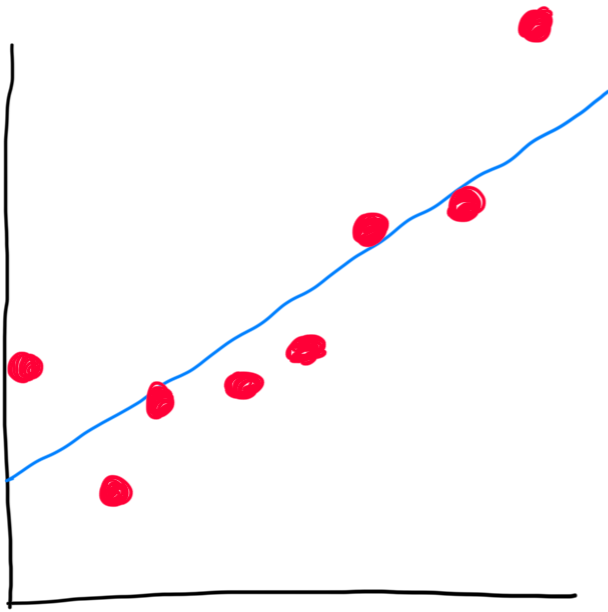
The dots marked on target (Bull's eye) then represent an individual realization of model based on training data

In certain cases, the dots will be densely positioned close to bull's eye ensuring that predictions made by model are close to actual data.

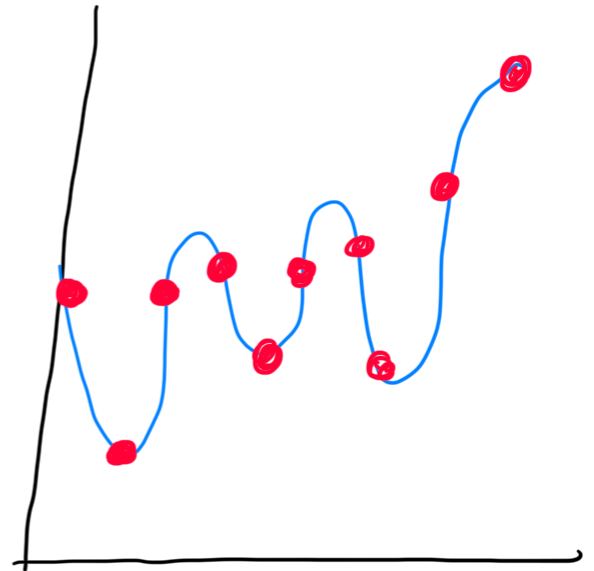
In other cases, the training data will be scattered across the target.

The more dots deviate from bull's eye,

the higher the bias & less accurate the model will be in predictive ability.



Underfitting
(Low variance
High Bias)



overfitting
(High Variance
Low Bias)